



LITERATURE REVIEW

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Literature review

1. Integration of security into the Software Development Lifecycle (SDLC)

Due to cyber threats becoming more sophisticated, integrating security into the Software Development Lifecycle (SDLC) has become a crucial priority in the modern software engineering. The conventional SDLC models have come under attack for delaying security concerns until far later on in the development process which has led to more vulnerabilities and more expensive cleanups (Dhuri, A., Samdani, K., Agwan, S., Wagh, S., Karande, Y., Narayan, Y., 2025).

On the other hand, current research highlights the use of Secure SDLC models which integrate security techniques like threat modelling, secure coding and continuous testing at every stage of development (Adeniyi, T., 2025). The literature generally agrees that early security integration greatly lowers system vulnerabilities. Research indicates that including security procedures throughout every stage of development enhances software resilience and lowers the risk of exploitation (Anasuri, S., 2024). This encourages software developers to switch from reactive to proactive security strategies.

2. DevSecOps as a modern security framework

By encouraging ongoing and automated security procedures, DevSecOps has become a leading methodology for incorporating security into the SDLC. By incorporating security measures into each phase of development and deployment, it expands on traditional DevOps. Automation, real-time vulnerability detection and continuous integration / continuous deployment (CI/CD) are important approaches (Jain, A., 2023).

DevSecOps increases security and development efficiency, according to numerous studies. DevSecOps for instance improves application security through automation and early risk identification according to a thorough literature research (Mohammed, K.I., Shanmugam, B. & El-Den, J., 2025). In the same manner, including security into DevOps procedures lowers security risks and boosts system resilience (ScienceDirect, 2025).

Nevertheless, despite its advantages research also points out difficulties in putting DevSecOps into practice. These include challenges integrating tools into current development pipelines, organizational opposition and lack of standardized processes (Mohammed, K.I., Shanmugam, B. & El-Den, J., 2025). This suggests that although DevSecOps is successful but adopting it is not simple.

3. Security challenges in the financial sector

Due to financial data being sensitive, the financial sector especially banks are prime targets for cyberattacks. According to recent studies, the use of microservices and cloud-native architectures has expanded the attack surface of banking systems increasing their susceptibility to security risks (ScienceDirect, 2025).

The literature generally agrees that in order to reduce these risks, financial organizations need to implement sophisticated security framework like DevSecOps and Zero Trust models. These methods provide for better threat detection ongoing monitoring and increased sensitive data protection (ScienceDirect, 2025). However other research argue that these frameworks' complexity could affect their implementation especially organizations with limited funding or experience.

4. Key themes and research gaps synthesized

A number of important themes reoccur throughout the literature. First of all, there is broad consensus that lowering vulnerabilities and raising software quality require incorporating security within the SDLC (OWASP, 2023; NIST, 2022). Second, frameworks like DevSecOps and Secure SDLC are commonly acknowledged as successful methods for accomplishing this integration (IBM, 2023; Mohammed, K.I., Shanmugam, B. & El-Den, J., 2025). Thirdly, because the financial industry is so risky it needs more security.

There are significant gaps in the literature regardless of these agreements. The majority of research concentrates on general software development environments and worldwide frameworks paying little attention to particular regional settings like South Africa (OWASP, 2023; NIST, 2022). Furthermore, despite the widespread promotion of DevSecOps, little research has been done on the real implementation issues it faces in banking institutions especially in developing countries (Mohammed, K.I., Shanmugam, B. & El-Den, J., 2025).

Therefore, by analysing how security practices are included into the SDLC within the South African banking industry and the difficulties involved in their implementation this study seeks to close those gaps (SABRIC, 2024).

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